Sol^n

North South University
Department of Electrical and Computer Engineering
Digital Logic Design CSE 231 Section-10
Quiz-1 **Spring 2026**

Name:
ID:

Total Marks-15 $L = 76 = (1001100)_2 = 1101010$ **Duration -20 minutes**

1 Convert the word DLD to its Gray equivalent. 3

$D = 68$, $L = \cancel{80}\ 76 \rightarrow \cancel{(1010000)_2 = (1111000)}_{gray}$.

$= (1000100)_2 = (1100110)_{gray}$

2 Consider the Boolean expression F(A, B, C, D) = Σ(1, 2, 3,4,14, 15). Minimize the expression using the sum of products (SOP) method. 5

$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + ABC\bar{D} + ABCD$.

$= \bar{A}\bar{B}D + \bar{A}\bar{B}C + \bar{A}B\bar{C}\bar{D} + ABC$ $\boxed{\text{Ans}}$

3 Write the maximum and minimum numbers representable in a 8-bit 2's complement system in binary, decimal and hexadecimal number system. 3

Max: $+127 = (01111111)_2 = (7F)_{16}$ Min $= -128 = (10000000)_2 = (80)_{H}$

4 Simplify the following Boolean expression: $\overline{(\bar{x}\bar{y} + z)} + z + xy + wz$ 4

$= \overline{\bar{x}\bar{y}} \cdot \bar{z} + z + xy + wz$.

$= (x+y)\bar{z} + z + xy + wz$

$= x\bar{z} + z + y\bar{z} + z + xy + wz$

$= (x+z) + (y+z) + xy + wz$

$= z + x(1+y) + wz + y = x + y + z$.

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